measured in anyone's reference frame), measure the total entropy of everything there. Then wait however long you might wish, and again measure the total entropy. If nothing has moved out through the "walls" of your region of space between the measurements, then the total entropy cannot have decreased, and it almost always will have increased, at least a little bit

What is this thing called "entropy" that increases? It is the amount of "randomness" in the chosen region of space, and the increase of entropy means that things are continually becoming more and more random.

Stated more precisely (see Box 12.3), *entropy is the logarithm of the number of ways that all the atoms and molecules in our chosen region can be distributed, without changing that region's macroscopic appearance.*[3] When there are many possible ways for the atoms and molecules to be distributed, there is a huge amount of microscopic randomness and the entropy is huge.

The law of entropy increase (the second law of thermodynamics) has great power. As an example, suppose that we have a room containing air and a few crumpled-up newspapers. The air and paper together contain less entropy than they would have if the paper were burned in the air to form carbon dioxide, water vapor, and a bit of ash. In other words, when the room contains the original air and paper, there are fewer ways that its molecules can be randomly distributed than when it contains the final air, carbon dioxide, water vapor, and ash. That is why the paper burns naturally and easily if a spark ignites it, and why the burning cannot easily and naturally be reversed to create paper from carbon dioxide, water, ash, and air. Entropy increases during burning; entropy would decrease during unburning; thus, burning occurs and unburning does not.

Stephen Hawking noticed immediately, in November 1970, the remarkable similarity between the second law of thermodynamics and his law of area increase, but it was obvious to him that the similarity was a mere coincidence. One would have to be crazy, or at least a little dim-witted, to claim that the area of a hole's horizon in some sense *is* the hole's entropy, Hawking thought. After all, there is nothing at all random about a black hole. A black hole is just the opposite of random; it is simplicity incarnate. Once a black hole has settled down into a quiescent state (by emitting